

Finite elliptic groups give rational solutions beyond the unit disk in arXiv:2403.11065

Candidate counterexample packet

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Status: candidate counterexample likely valid. This is a full negative answer to the first open question as stated, while the non-elementary/lattice-generated variant remains open.

1 Source question

For a probability measure μ on $PSU(1, 1)$, the source studies holomorphic solutions of

$$\int f(\gamma^{-1}.z)(\gamma^{-1})'(z) d\mu(\gamma) - f(z) = \int \frac{d\mu(\gamma)}{z - \gamma.\infty}. \quad (1)$$

Its first open question asks whether, for finitely supported μ , every solution has radius of convergence exactly 1. It also proposes that rational solutions should occur only when μ is supported on a single element [1].

6 Open questions

- It is easy to see from the proof of Corollary 1.11 that we actually get non-existence of solutions $f(z) = \sum a_{k+1}z^k$ to (13) with $\limsup_{n \rightarrow \infty} |a_k|^{1/k} < \varepsilon$ for some small ε , as only one preimage of the chosen contour explodes, so we can bound the radius of the convergence of the solution. Ideally, one would like to show that for finitely supported μ every solution of (13) has radius of convergence exactly 1. Keep in mind that this result would almost close the smoothness gap: it is known that absolutely continuous densities stationary with respect to finitely supported measures can belong to $C^n(S^1)$ for any $n > 1$.

The Douglas-Shields-Shapiro theorem implies that any holomorphic function with the radius of convergence exceeding 1 is either cyclic with respect to the backward shift or rational. It is reasonable to assume that (13) only has rational solutions when μ is supported on a single element, and we conjecture that former never happens.

The complete first bullet of Section 6 in the latest arXiv source, page 19.

2 Counterexample

Theorem 1. *There is a two-point probability measure μ on $PSU(1, 1)$, both of whose support elements move the origin, for which (1) has the nonzero rational solution*

$$f(z) = \frac{1}{2 - z}.$$

Its Taylor series at the origin has radius of convergence 2. Consequently, both literal assertions in the source question are false.

Proof. Put $a = \frac{1}{2}$, let $\omega = e^{2\pi i/3}$, and define

$$h(z) = \frac{z+a}{1+az}, \quad r_j(z) = \omega^j z, \quad \gamma_j = h \circ r_j \circ h^{-1} \quad (j = 1, 2).$$

These are elements of $PSU(1, 1)$. Let

$$\mu = \frac{1}{2}(\delta_{\gamma_1} + \delta_{\gamma_2}), \quad \nu = h_* m,$$

where m is normalized Lebesgue measure on $\mathbb{T}$. Rotation invariance gives $(r_j)_* m = m$, and hence

$$(\gamma_j)_* \nu = h_*(r_j)_* m = h_* m = \nu.$$

Thus ν is invariant under each support element and, in particular, is μ -stationary.

We compute its Cauchy transform. For $k \geq 1$, the mean-value property for the holomorphic function h^k gives

$$\int_{\mathbb{T}} \xi^k d\nu(\xi) = \int_{\mathbb{T}} h(\eta)^k dm(\eta) = h(0)^k = a^k.$$

Since a is real and $|\xi| = 1$, conjugation yields $\int \xi^{-k} d\nu(\xi) = a^k$. Therefore, for $|z| < 1$,

$$f_\nu(z) = \int_{\mathbb{T}} \frac{d\nu(\xi)}{\xi - z} = \sum_{n \geq 0} z^n \int_{\mathbb{T}} \xi^{-(n+1)} d\nu(\xi) = \sum_{n \geq 0} a^{n+1} z^n = \frac{a}{1 - az} = \frac{1}{2 - z}.$$

The source's main theorem says that the Cauchy transform of every stationary probability measure satisfies (1). Hence f_ν is a solution. It is nonzero and rational, and its unique pole is at $z = 2$, so its Taylor radius is 2.

Finally, for $j = 1, 2$,

$$\gamma_j(0) = h(-a\omega^j) = \frac{a(1 - \omega^j)}{1 - a^2\omega^j} \neq 0.$$

The two maps are distinct, so $|\text{supp } \mu| = 2$. This completes all claimed properties. $\square$

3 Why this is the correct obstruction

The example is not an accident at the level of a single formula. If K is any finite rotation group, $h \in PSU(1, 1)$, and μ is supported on at least two elements of hKh^{-1} , then $h_* m$ is invariant under every support element. When $h(0) = a$, after a harmless rotation making $a > 0$, its Cauchy transform is $a/(1 - az)$ and has radius $1/a > 1$. Thus an entire family of finite elliptic examples defeats the proposed conclusion.

At the un-conjugated center the kernel is also transparent. For $\rho_\alpha(z) = \alpha z$,

$$f(z) = \sum_{n \geq 0} c_n z^n \implies f(\rho_\alpha^{-1} z)(\rho_\alpha^{-1})'(z) = \sum_{n \geq 0} c_n \alpha^{-(n+1)} z^n.$$

For a probability measure on rotations, a coefficient c_n may be nonzero exactly when every supported rotation satisfies $\alpha^{n+1} = 1$. In a cyclic group of order q , all functions $z^{q-1}g(z^q)$ solve the homogeneous equation. Choosing g entire gives further nonzero entire solutions with multi-point support. Conjugation produces the stronger off-center rational solution used in Theorem 1.

4 Verification and scope

The accompanying SymPy script performs an independent exact check for $a = 1/2$ and $q = 3$: it constructs γ_1, γ_2 and their inverses, checks both maps move 0, and simplifies the left side of (1) minus the right side to zero. All arithmetic lies in $\mathbb{Q}(i\sqrt{3})(z)$; numerical approximation is not used.

This counterexample meets the source’s earlier nontriviality condition that some supported map not fix the origin. It does generate only a finite elliptic (hence elementary) subgroup. It therefore does not settle a strengthened question requiring the support to generate a non-elementary group or a lattice, which is the setting relevant to the singularity conjecture. Such a hypothesis, or at least exclusion of compact elementary subgroups, is necessary for the proposed rigidity statement.

5 Bounded novelty check

The search through 11 August 2026 covered the run’s registry, solution, attempt, and proof-gap indexes; arXiv:2403.11065v2 (17 April 2025); the July 2026 Journal of Functional Analysis publication page; the exact open-question phrases; and combinations of the title or identifier with *rotation*, *rational solution*, and *radius of convergence*. It found no later source explicitly recording this counterexample. Novelty confidence is moderate pending specialist review; the exact computation is independent of novelty.

6 Conclusion

The first open question has a negative answer as written. A two-point measure supported on an off-center finite elliptic group admits the stationary Cauchy transform $1/(2 - z)$, whose radius is 2. The same example refutes the proposed one-point-support restriction for rational solutions. Human review should confirm the literal scope and, if desired, restate the surviving problem with a non-elementary or lattice-generation assumption.

References

- [1] P. Kosenko, *On a complex-analytic approach to stationary measures on S^1 with respect to the action of $PSU(1, 1)$* , arXiv:2403.11065v2; J. Funct. Anal. 291 (2026), 111476. <https://arxiv.org/abs/2403.11065>.